

Lab sheet 12**Submission Due: 02.10.2024****Title:** Introduction to PHP framework (Laravel)**Aims:**

- Setting up development environment
- Database connectivity
- Project creation
- Understanding the Laravel's core concepts
- Project structure navigation

Tasks:

- Install PHP
- Install Laravel and Composer
- Create a Laravel project
- Navigate through Laravel documentation and architecture
- Introduce Laravel database
- Review of Laravel file structure

Activities :**1. Install PHP**

On Linux and macOS, PHP is typically pre-installed. To check the version of PHP installed, run the following command:

php -v

If PHP is not installed, you can install it using your package manager.
On *Ubuntu*, you can install PHP with the following command:

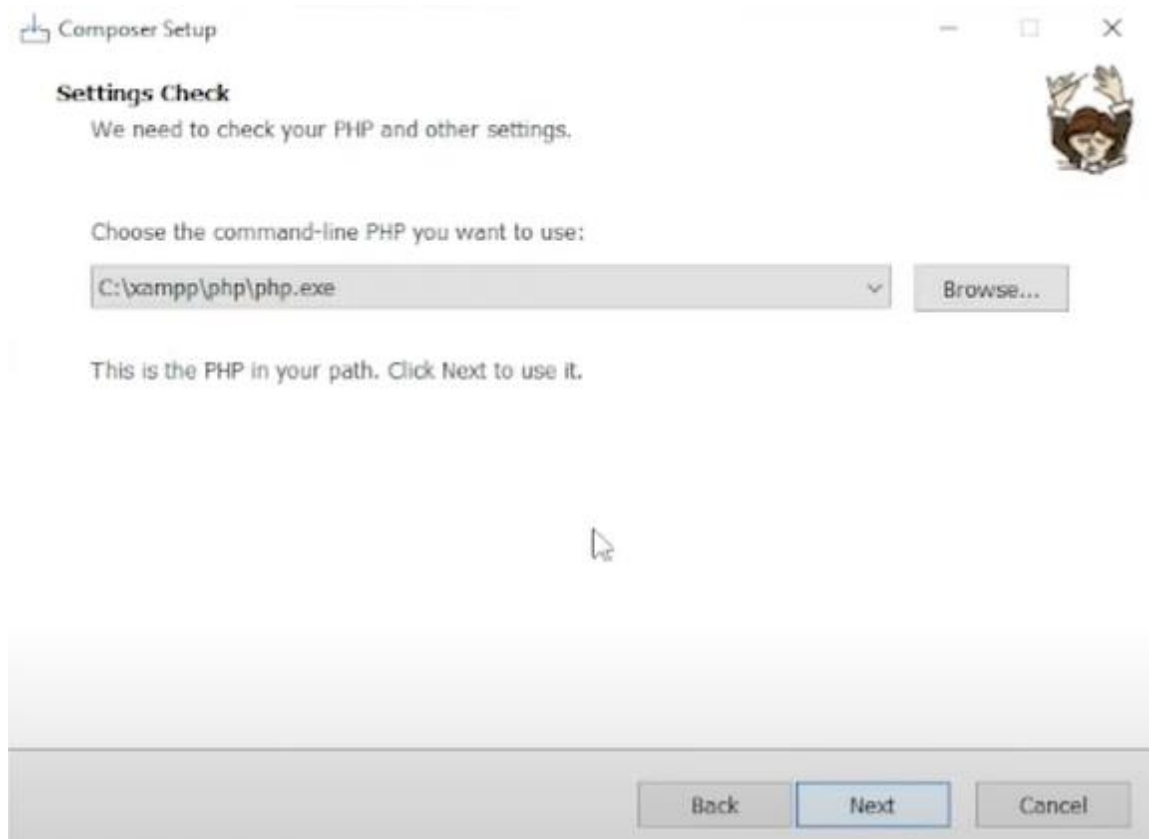
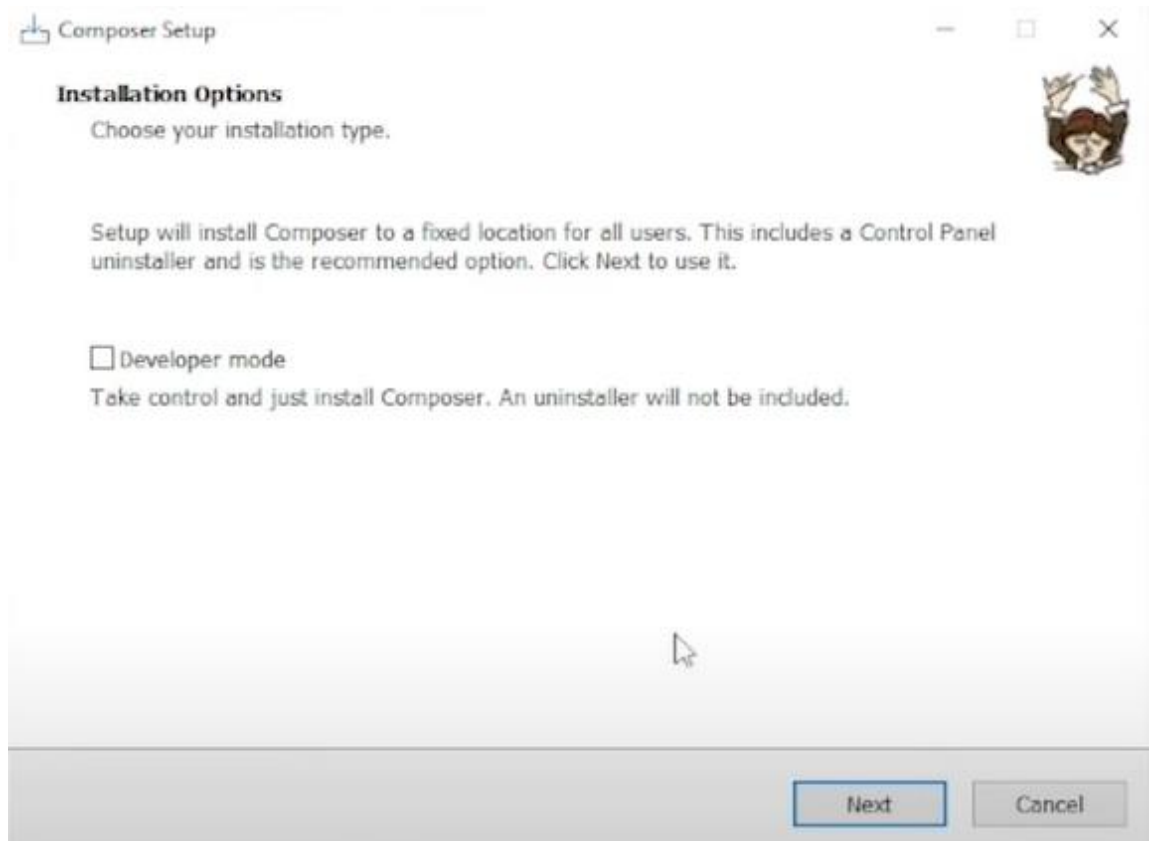
sudo apt install php

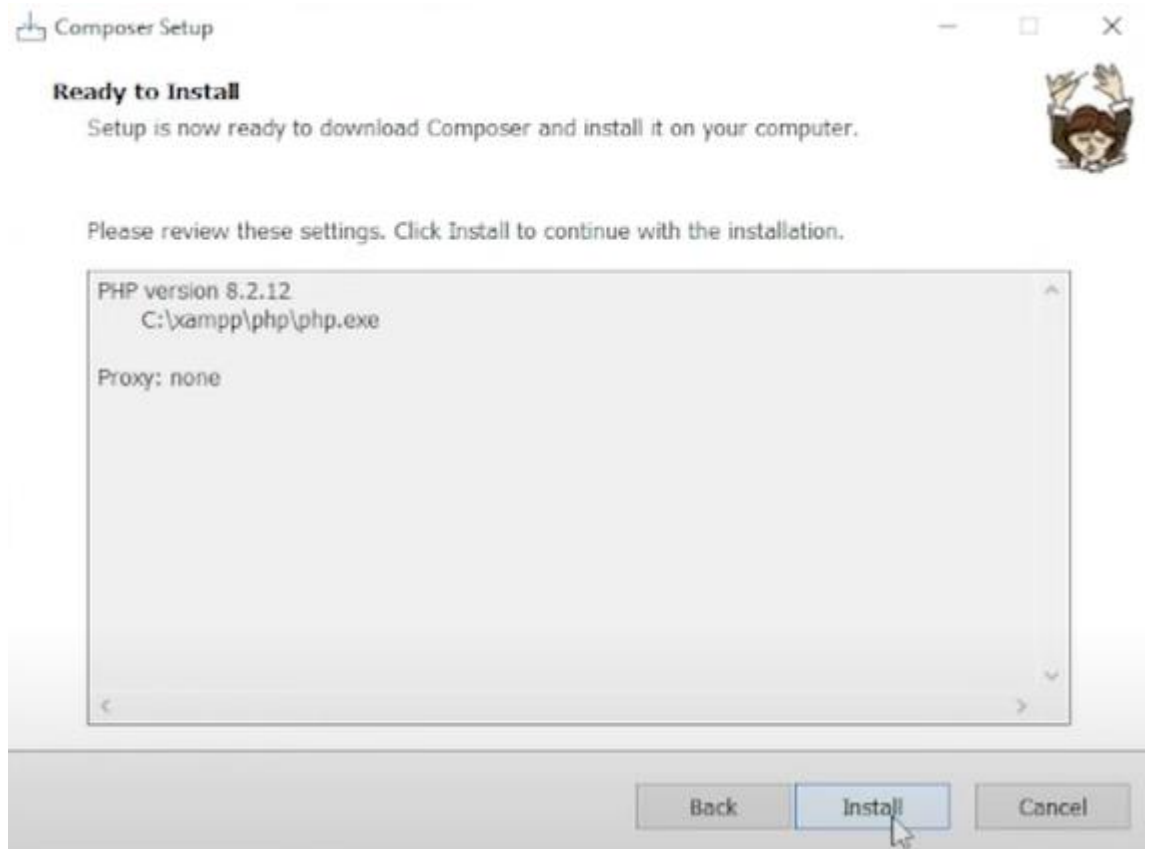
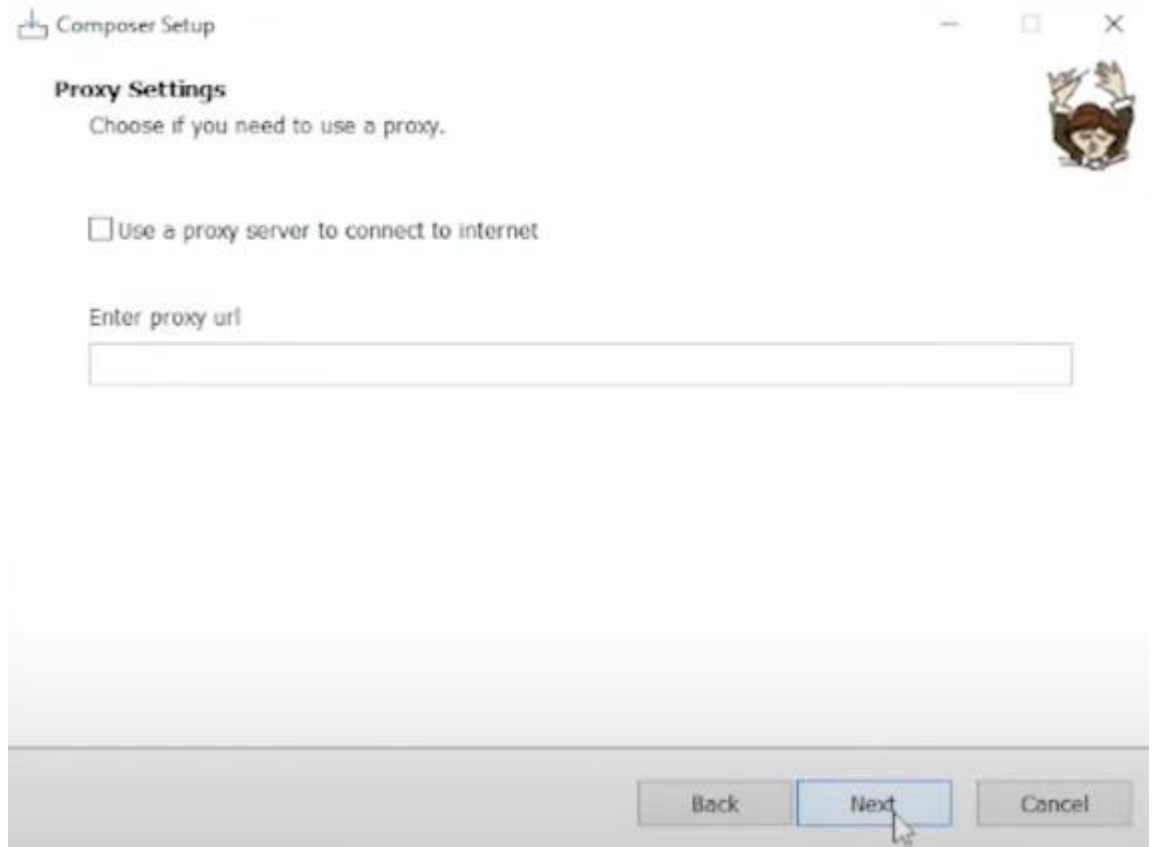
On *Windows*, You can refer this documentation :

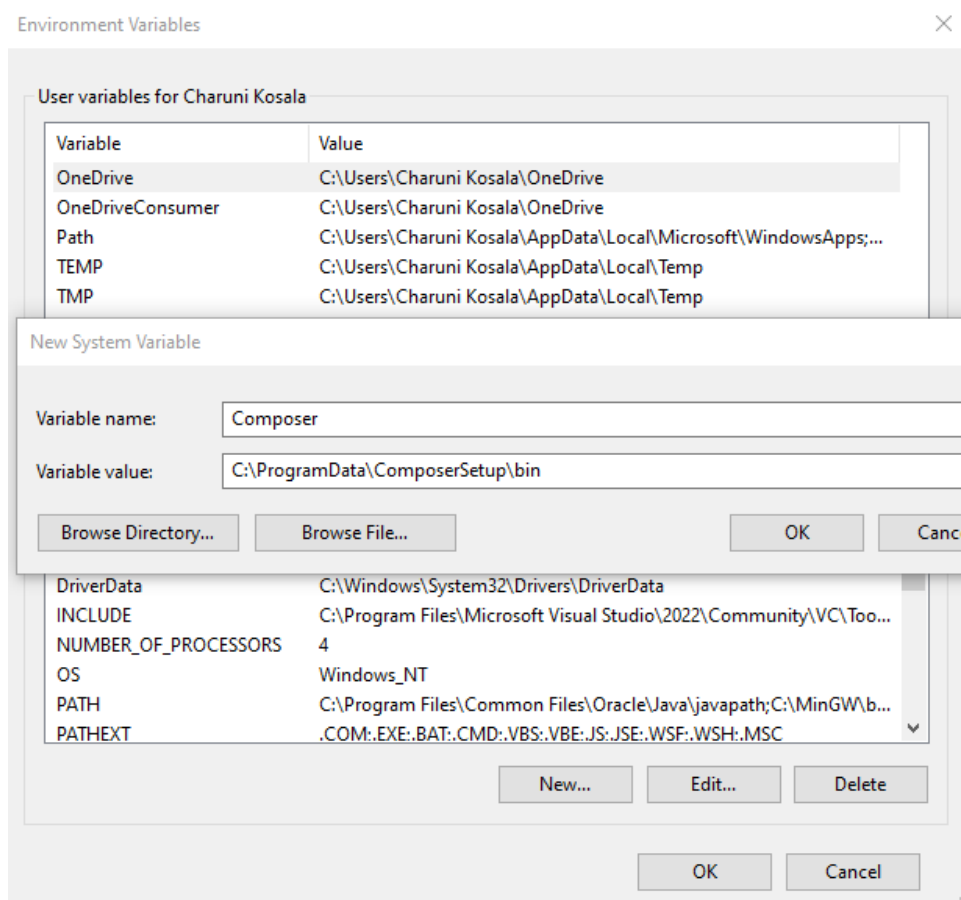
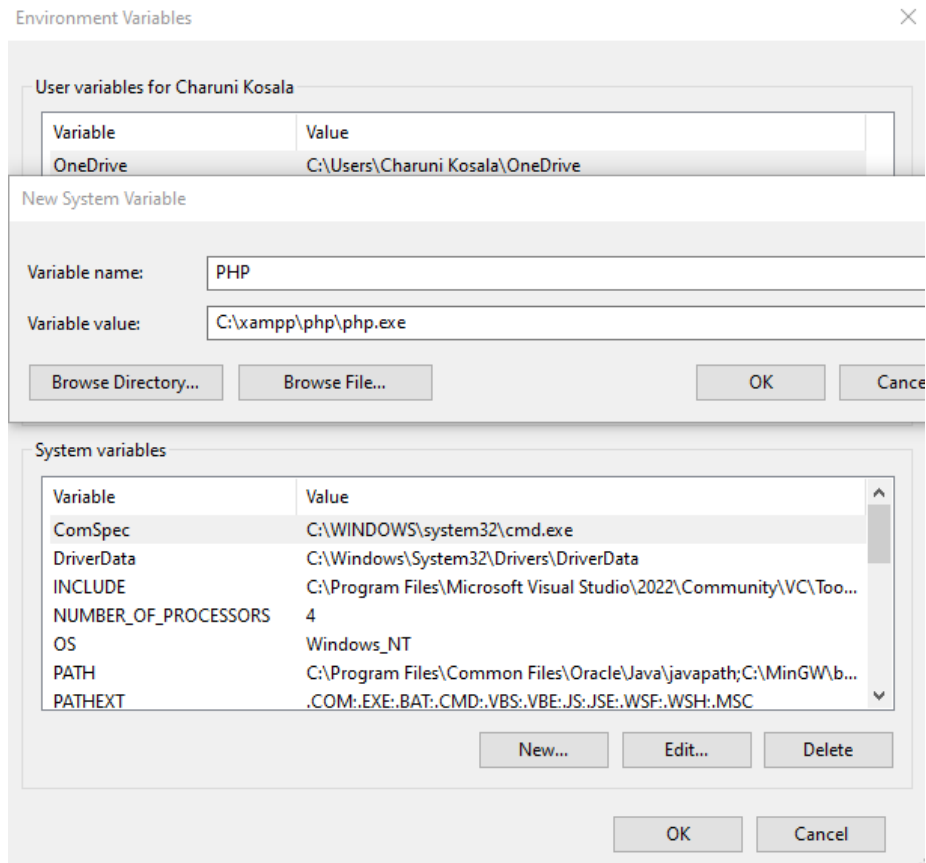
<https://windows.php.net/download>

2. Install Laravel and Composer

<https://getcomposer.org/doc/00-intro.md>







. To check the version of PHP installed, run the following command:

composer --version

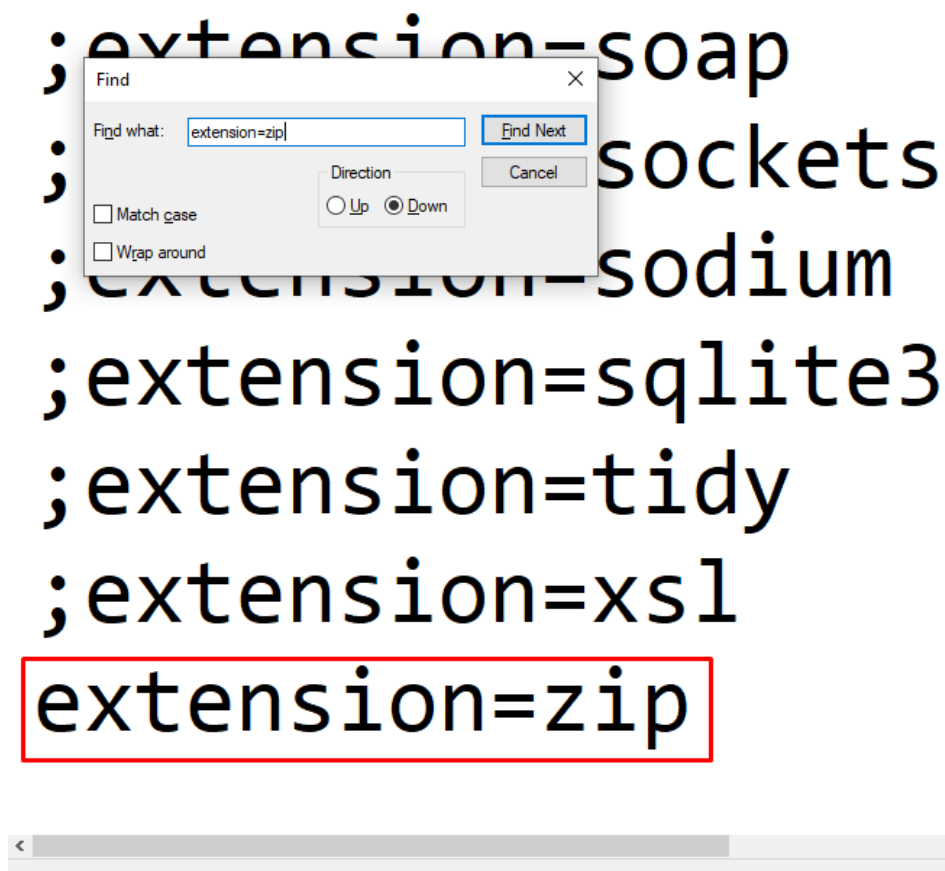
3. Install Laravel

To install Laravel, run the following command:

composer global require laravel/installer

Do these changes in php.ini file

php - Notepad
File Edit Format View Help

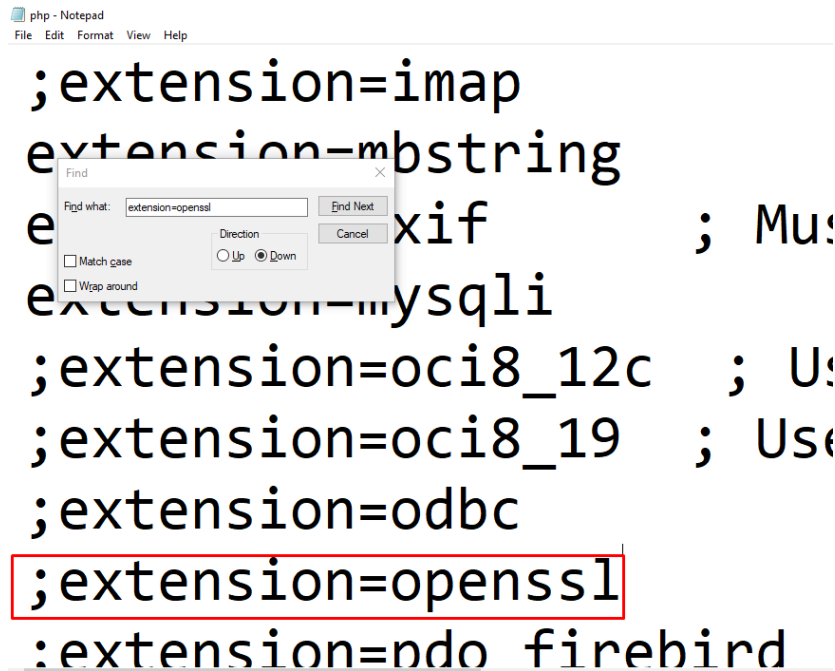


php - Notepad
File Edit Format View Help

```
; See https://github.com/php/extension-ldap  
extension=ldap  
extension=curl  
extension=ffi  
extension=ftp  
extension=fileinfo  
extension=gd  
extension=gettext  
extension=intl  
extension=imap  
extension=mbstring  
extension=exif  
extension=mysqli  
extension=oci8_12c  
extension=oci8_19  
extension=odbc  
extension=openssl
```

Find
Find what: extension=file
Find Next
Cancel
Direction
☒ Up ☐ Down
☐ Match case
☐ Wrap around

Find
Find what: extension=mysql
Find Next
Cancel
Direction
☐ Up ☒ Down
☐ Match case
☐ Wrap around



4. Create Laravel project

Use this command :

```
laravel new example-app
```

Run the project :

```
cd example-app
```

```
php artisan serve
```

5. Navigate through Laravel documentation and architecture

The Laravel documentation is a comprehensive resource for learning about all aspects of the Laravel framework. It can be found at the following URL:

<https://laravel.com/docs/11.x>

The documentation is divided into several sections, including:

- *Getting Started*: This section covers the basics of Laravel, such as setting up a new project, routing, and controllers.
- *Views*: This section covers how to create and render views in Laravel.
- *Blade*: This section covers the Blade templating engine, which is used to create views in Laravel.
- *Artisan*: This section covers the Artisan console, which provides a number of helpful tools for managing Laravel applications.
- *Eloquent*: This section covers Eloquent, the Laravel ORM, which makes it easy to interact with databases.
- *Authentication and Authorization*: This section covers how to implement authentication and authorization in Laravel applications.

- *Testing*: This section covers how to write and run tests for Laravel applications.
- *Deployment*: This section covers how to deploy Laravel applications to production servers.

Architecture:

Laravel is a full-stack web framework, meaning that it provides all of the tools needed to develop web applications including:

- *MVC architecture*: Laravel uses the MVC architecture, which separates the application into three layers: models views, and controllers. This makes the code more modular and easier to maintain.
- *Routing*: Laravel provides a robust routing system that allows you to define how your application responds to different HTTP requests.
- *Templating*: Laravel uses the Blade templating engine, which is a powerful and flexible way to create views.
- *Database abstraction layer*: Laravel provides Eloquent, the Laravel ORM, which makes it easy to interact with databases.
- *Authentication and authorization*: Laravel provides built-in support for authentication and authorization, making it easy to secure your applications.
- *Testing framework*: Laravel provides a built-in testing framework that makes it easy to write and run tests for your applications.

6. Introduce Laravel Database

<https://laravel.com/docs/11.x/eloquent>

php artisan make:model Employee -m

```
public function up()
{
    Schema::create('employees', function (Blueprint $table) {
        $table->id();
        $table->string('first_name');
        $table->string('last_name');
        $table->string('email')->unique();
        $table->string('phone')->nullable();
        $table->string('position');
        $table->decimal('salary', 8, 2);
        $table->date('hired_date');
        $table->timestamps();
    });
}
```

Discussion:

- MVC architecture
- object-relational mapper (ORM)